

Mirza Usama Ali Baig

Senior Full-Stack Engineer · NestJS · Laravel · FastAPI · Next.js / React · Nuxt · AI Integrations

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SUMMARY

Senior full-stack engineer with 4 years of experience building and shipping production SaaS: executive-talent, logistics dispatch, workforce time-tracking, tax-practice management, healthcare EHR integration and private-aviation platforms, several of them live and linked below. I build backends in NestJS, Laravel and Python FastAPI behind React/Next.js and Nuxt frontends, with multi-tenant security (RBAC, MFA, IP allowlisting), real-time systems (Socket.IO, queues), Stripe billing, and document pipelines with OpenAI/LangChain and OCR. I own features end to end, from schema design to CI/CD and deployment on AWS.

EXPERIENCE

MERN Stack Developer — Code Upscale

Jun 2025 – Present

- Built and led the backends of production products including ExecSuccession, On Time Transport and the Vitals 360 Kipu EHR sync, and reviewed and merged 200+ team pull requests across client repos.
- Worked across the Laravel 12 backend and Electron desktop agent of TrackFlow, and handled security audits and malware-injection clean-ups on client codebases (ArchiTAXture, On Time Transport).
- Built secure multi-tenant platforms (JWT, RBAC, MFA/2FA, privileged-access controls, IP allowlisting) and AI document pipelines with OpenAI, LangChain, Tesseract OCR, MRZ parsing and PDF/DOCX extraction.
- Built Stripe subscription billing, S3 presigned-URL document flows and Socket.IO real-time dashboards backed by Redis/BullMQ, across MongoDB and PostgreSQL/MySQL stacks.

Full Stack Developer — Aug AI

Oct 2024 – May 2025

- Built Socket.IO real-time experiences and conversational AI workflows, including the Safeli messenger backend (chatrooms, projects, pinned and read messages).
- Delivered clean-architecture NestJS systems in strict TypeScript, with auth guards, interceptors, Swagger API docs and Docker-based deployments.

Associate Software Engineer — Conovo Technologies

Aug 2023 – Sep 2024

- Built MERN/NestJS backends, REST and GraphQL APIs and live Socket.IO features, with Passport.js and JWT authentication and authorization.

Machine Learning Engineer — Cyberautix Technologies Limited

Feb 2023 – Aug 2023

- Built YOLO computer-vision pipelines for real-time object detection, including a diamond classification and labelling system that grades stones by value across images and video, served through Python FastAPI inference APIs.
- Prepared and augmented large image datasets, trained models and tuned hyperparameters for production stability.

Associate Software Engineer — Hi Tech Provider

Nov 2022 – Jul 2023

- Built Python and JavaScript backend services, Puppeteer scraping workflows and secure Express.js APIs, plus ResNet-50 image-classification and feature-extraction solutions.

SELECTED PROJECTS

ExecSuccession / ExecExtend — AI Executive Succession & Offer Platform

Jun 2025 – Present

Live: execsuccession.com · execextend.com · Founding backend engineer

NestJS · MongoDB · Next.js 15 · OpenAI · Stripe · BullMQ/Redis · AWS S3 · Puppeteer

- Wrote the first commit and led the backend (about 67% of commits in its first 6 months), growing it to roughly 45 NestJS modules for HR leaders and boards.
- Built the auth, candidate/S3 document, assessment-engine (MCQ, rating and text; public submission APIs) and succession-plan modules, with 4 readiness tiers and 1–5 competency frameworks.
- Added OpenAI resume analysis and AI-generated competency summaries; implemented Stripe subscriptions with webhook reconciliation, multi-tenant RBAC with MFA/2FA and IP allowlisting, audit logs and board-ready PDF reports. Also contributed to ExecExtend (offer optimisation, pay-equity scoring).

On Time Transport — Trucking Logistics & Dispatch Platform

Jul 2026 – Present

Live: [On Time Transport dispatch board](#) · Lead contributor, backend and web

NestJS 11 · PostgreSQL/TypeORM · Redis · BullMQ · Socket.IO · React 19 · Vite · Tailwind v4 · AWS S3

- Top contributor on backend and web (about 55% and 74% of commits): 13 domain modules (loads, drivers, vehicles, plants, customers, chat, notifications, imports, roles) in a layered clean architecture, with portals for admin, ops, dispatch, billing and driver managers.
- Built a drag-and-drop dispatch board with multi-trip load groups, open-to-claim loads and rolled-up partial-delivery status.
- Integrated the Samsara fleet-telematics API, built a generic CSV import engine on BullMQ workers, and added truss delivery-schedule PDF parsing and import with race-free duplicate checks.

TrackFlow — Hubstaff-style Time-Tracking & HR/Payroll SaaS

Jul 2026 – Present

Live: trackflow.codeupscale.com · Multi-tenant · Laravel backend and Electron desktop agent

Laravel 12 · PostgreSQL · Redis/Horizon · Reverb WebSockets · Sanctum · PHPUnit · Next.js 16 · Electron · SQLite · Stripe

- Laravel backend: made worked-time totals consistent across dashboard, timer and every report, closed superseded sessions at their last input, anchored overnight shifts correctly, added HR attendance rules and RBAC role seeding, and hardened PHPUnit tests against timezone flakes.
- Shipped the first local-first desktop client with idle-detection fixes, minimum-version enforcement and Linux/Wayland screen capture over PipeWire; built the desktop release CI in GitHub Actions.

ArchiTAXture / WeCloseBooks — Tax Practice Management SaaS

Jun 2026 – Present

Live: app.weclosebooks.com · Codebase audit, security and review

NestJS 11 · TypeORM/PostgreSQL · Redis · Anthropic Claude · AWS S3 · Next.js 15 · RTK Query · Tailwind · Radix UI

- Audited the inherited platform (~400 files, ~70 tables, 100+ endpoints) and wrote a phased hardening plan covering IDOR gaps, unsigned webhooks, missing RBAC guards and tests.
- Found and removed malicious code injected into staging; reviewed and merged rebuild features (client tax profiles and dependents, role-permission counts, auth screens, client-detail editing).

UrgentFlights — Private-Aviation Charter Platform

Nov 2025 – Jan 2026

Live: app.urgentflights.com · urgentflights.com

Nuxt 3 · Vue 3 · MongoDB · OpenAI · Tesseract OCR · AWS S3 · SendGrid · Twilio · Socket.IO

- Owned the AI email-to-quote pipeline: an IMAP inbox is fetched, PDF attachments are parsed with several fallback parsers and OCR, OpenAI extracts the fields, and draft quotes are created automatically.
- Built a flight matcher with multi-criteria scoring, confidence ratings and date flexibility, and the client portal (flight requests, quotes, passenger ledger) with RBAC and two-factor auth.

Vitals 360 — Kipu EHR Sync Service (Healthcare)

Jul 2025 – Aug 2025

Live: portal.vitals-360.com · Sole author of the sync service

NestJS 11 · PostgreSQL/TypeORM · @nestjs/schedule · Kipu REST API · Winston · ExcelJS

- Daemon syncing Kipu EHR patients into the Vitals 360 clinic-monitoring platform with diff-based change detection; room/occupancy mapping with Excel import, sync-dashboard API, webhooks and admin users.

Smart Visa Assistant — Visa Application Automation

Mar 2026 – Present

NestJS · TypeORM/MySQL · Redis/Bull · Playwright · Tesseract OCR · OpenAI · AWS S3

- Visa-automation platform that validates passports, parses forms and captures structured immigration data from documents; Playwright submits government visa forms with OTP relay and completion reporting.

Wrstopia — Luxury Watch Marketplace

Jul 2025 – Aug 2025

NestJS · MongoDB/Mongoose · JWT · Multer · Nodemailer

- Sole backend engineer: consumers submit sell/service requests and dealers bid; built per-role auth, the bidding engine, a revenue dashboard and admin tooling.

MIHU — Furniture Marketplace with Visual Search (Final-Year Project)

Oct 2022 – Jun 2023

Live: mihu-prod.vercel.app · Team of 4

Python · FastAPI · TensorFlow ResNet-50 · FAISS · OpenCV · Puppeteer · Express · MongoDB · Stripe · React · Redux Toolkit

- Multi-vendor furniture marketplace with seller and customer logins. Built the Puppeteer scrapers for six furniture brands and the "search by photo" engine: ResNet-50 features, FAISS similarity search, served over FastAPI.

Earlier Projects

2023 – 2025

- SALM** (NestJS, PostgreSQL, OpenAI): streaming/batch pipelines for high-volume transactional data, Swagger partner APIs, AI document analysis. **PUNT** (NestJS, MongoDB, Socket.IO): migrated a betting platform off Firebase with live odds and cron payouts. **Alchemist** (NestJS, MongoDB): valuation, ID-verification and background-check integrations with PDF reports.
- Personal:** Metro (Nuxt 4 + MongoDB e-commerce with admin/seller/customer roles), Committee System (NestJS + Next.js, live), Kafka event-driven NestJS app, Stripe event-management API, Gemini AI chatbot (Next.js + Firebase).

SKILLS	
Backend	NestJS, Node.js, Express, TypeScript, Laravel 12 (Horizon, Reverb, Sanctum), Python, FastAPI, GraphQL, Swagger
Frontend	React, Next.js, Nuxt/Vue 3, Angular, Vite, Redux Toolkit, TanStack Query, Tailwind CSS, Electron
Data & Queues	PostgreSQL, MySQL, MongoDB, TypeORM, Mongoose, Sequelize, Redis, BullMQ, Kafka, SQLite
AI & Automation	OpenAI, LangChain, Anthropic Claude, Google Gemini, Tesseract OCR, PDF/DOCX extraction, Playwright, Puppeteer, TensorFlow, YOLO, ResNet-50, FAISS
Security & Payments	JWT, RBAC, MFA/2FA, IP allowlisting, Passport.js, Stripe subscriptions and webhooks
Cloud & DevOps	AWS (S3, Amplify), Vercel, Firebase, Docker, GitHub Actions CI/CD, Nginx, PHPUnit, Jest

EDUCATION, CERTIFICATIONS & LANGUAGES

BSc Computer Science, University of Central Punjab, Lahore (CGPA 3.84; FYP: MIHU) · Intermediate in CS, GCU Lahore · Languages: English (Fluent), Urdu (Native)

Certifications: JavaScript Algorithms & Data Structures (freeCodeCamp) · Ruby on Rails Boot Camp (Devsinc) · Artificial Intelligence (Samsung Innovation Campus) · Python for Everybody & Python Data Structures (Coursera)